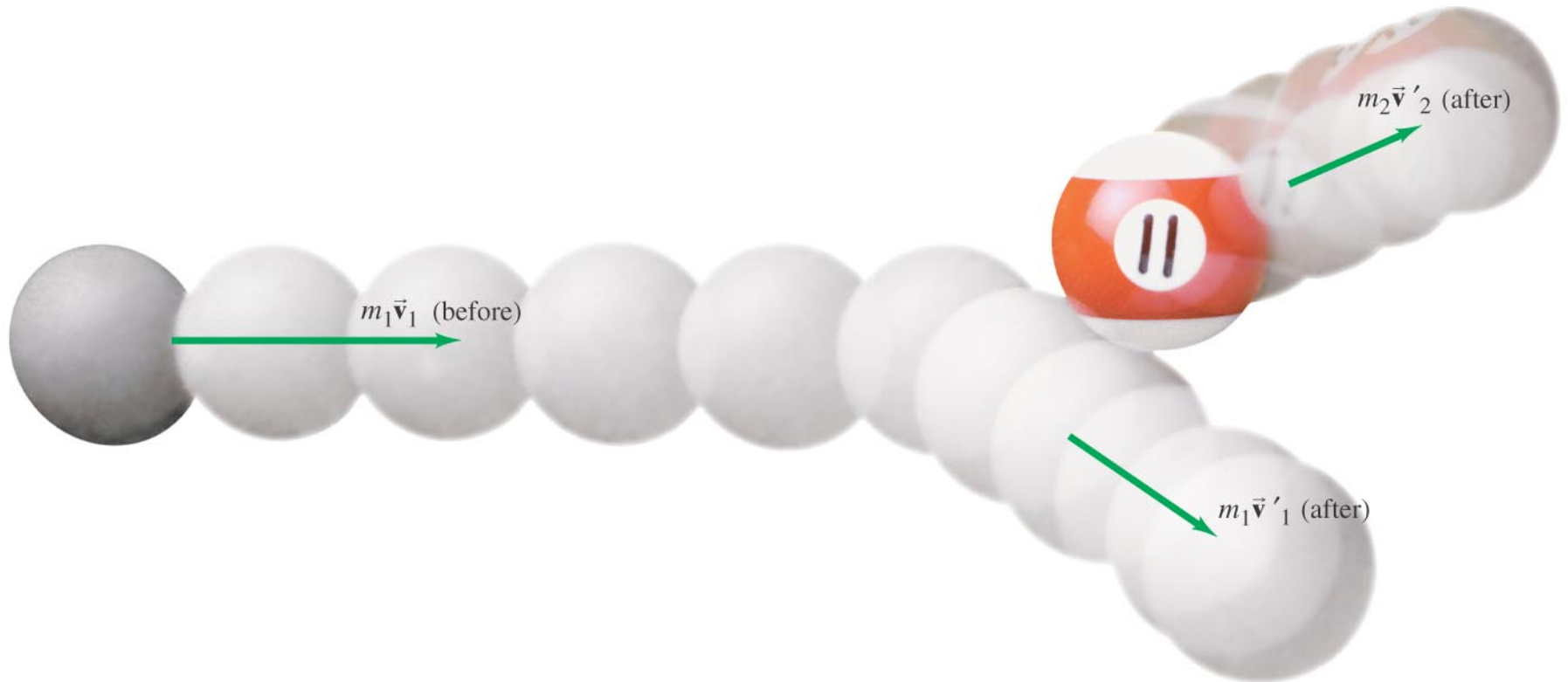


Chapter 9

Linear Momentum



Units of Chapter 9

- **Momentum and Its Relation to Force**
- **Conservation of Momentum**
- **Collisions and Impulse**
- **Conservation of Energy and Momentum in Collisions**
- **Elastic Collisions in One Dimension**

Units of Chapter 9

- **Inelastic Collisions**
- **Collisions in Two or Three Dimensions**
- **Center of Mass (CM)**
- **Center of Mass and Translational Motion**
- **Systems of Variable Mass; Rocket Propulsion**

9-1 Momentum and Its Relation to Force

Momentum is a vector symbolized by the symbol $\vec{p}$, and is defined as

$$\vec{p} = m\vec{v}.$$

The rate of change of momentum is equal to the net force:

$$\Sigma \vec{F} = \frac{d\vec{p}}{dt}.$$

This can be shown using Newton's second law.

9-1 Momentum and Its Relation to Force

Example 9-1: Force of a tennis serve.



For a top player, a tennis ball may leave the racket on the serve with a speed of 55 m/s (about 120 mi/h). If the ball has a mass of 0.060 kg and is in contact with the racket for about 4 ms (4×10^{-3} s), estimate the average force on the ball. Would this force be large enough to lift a 60-kg person?

9-1 Momentum and Its Relation to Force

Example 9-2: Washing a car: momentum change and force.

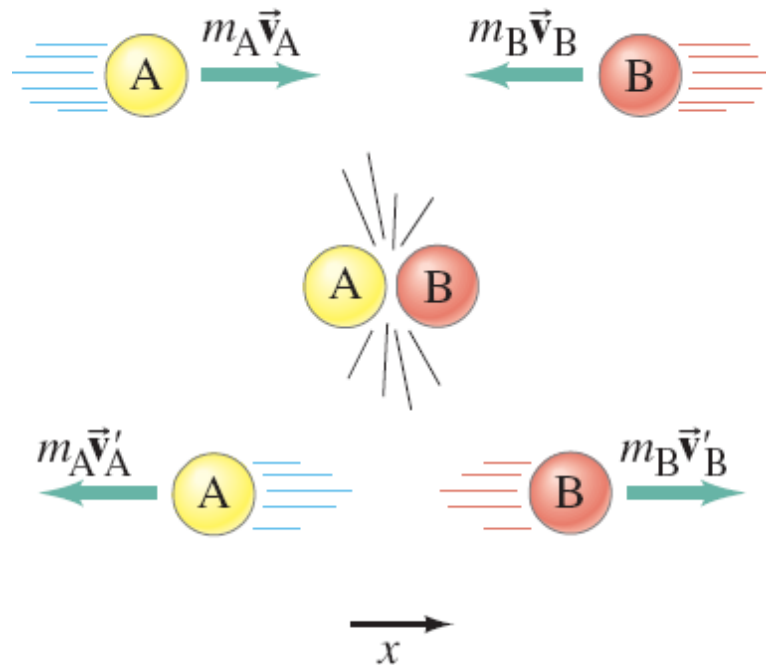
Water leaves a hose at a rate of 1.5 kg/s with a speed of 20 m/s and is aimed at the side of a car, which stops it. (That is, we ignore any splashing back.) What is the force exerted by the water on the car?



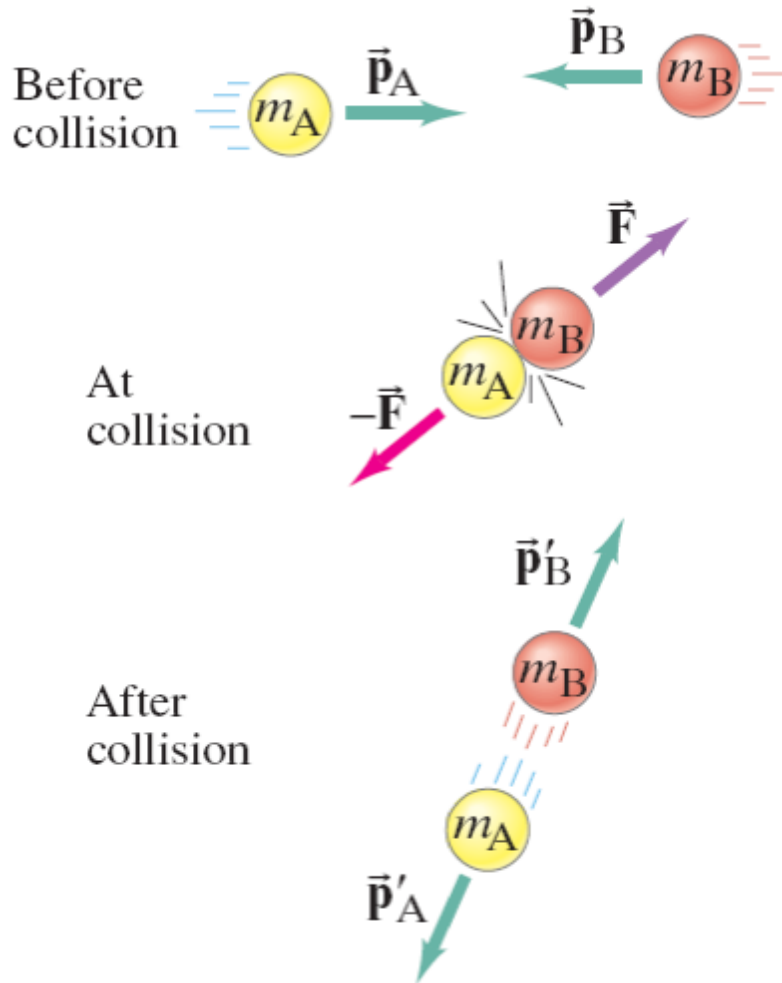
9-2 Conservation of Momentum

During a **collision**, measurements show that the **total momentum does not change**:

$$m_A \vec{v}_A + m_B \vec{v}_B = m_A \vec{v}'_A + m_B \vec{v}'_B.$$



9-2 Conservation of Momentum



Conservation of momentum can also be derived from Newton's laws. A collision takes a short enough time that we can ignore external forces. Since the internal forces are equal and opposite, the total momentum is constant.

9-2 Conservation of Momentum

For more than two objects,

$$\frac{d\vec{\mathbf{P}}}{dt} = \sum \frac{d\vec{\mathbf{p}}_i}{dt} = \sum \vec{\mathbf{F}}_i.$$

Or, since the internal forces cancel,

$$\frac{d\vec{\mathbf{P}}}{dt} = \sum \vec{\mathbf{F}}_{\text{ext}}.$$

9-2 Conservation of Momentum

This is the law of conservation of linear momentum:

when the net external force on a system of objects is zero, the total momentum of the system remains constant.

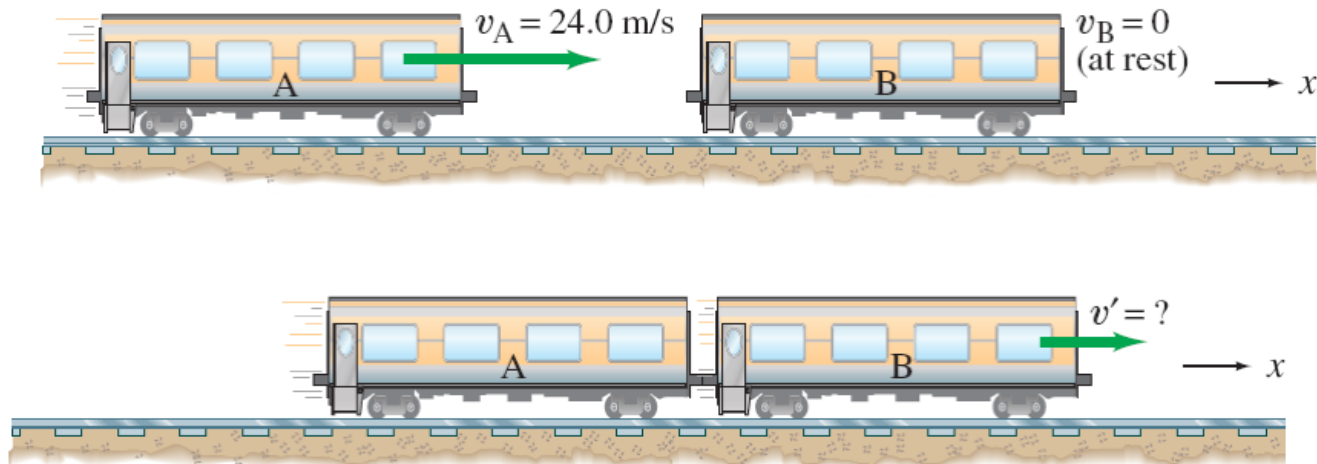
Equivalently,

the total momentum of an isolated system remains constant.

9-2 Conservation of Momentum

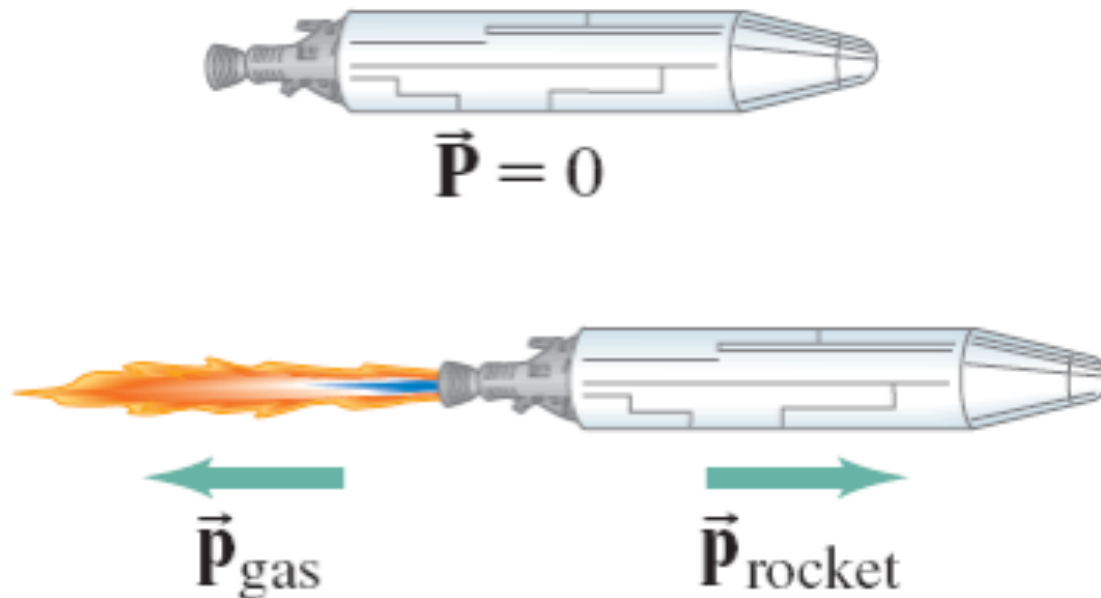
Example 9-3: Railroad cars collide: momentum conserved.

A 10,000-kg railroad car, A, traveling at a speed of 24.0 m/s strikes an identical car, B, at rest. If the cars lock together as a result of the collision, what is their common speed immediately after the collision?



9-2 Conservation of Momentum

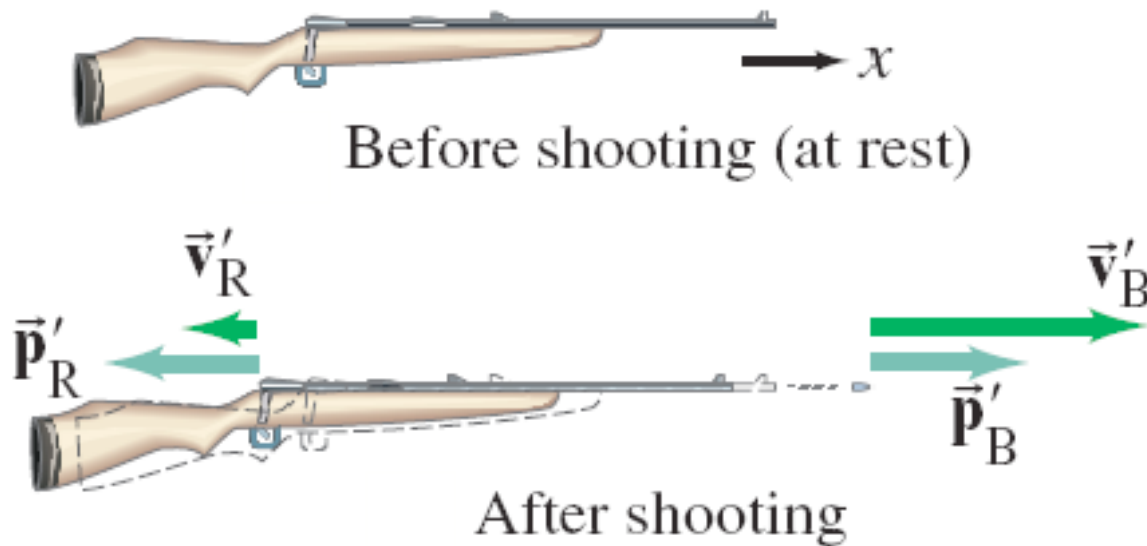
Momentum conservation works for a **rocket** as long as we consider the rocket and its **fuel** to be one system, and account for the **mass loss** of the rocket.



9-2 Conservation of Momentum

Example 9-4: Rifle recoil.

Calculate the recoil velocity of a 5.0-kg rifle that shoots a 0.020-kg bullet at a speed of 620 m/s.



9-3 Collisions and Impulse

This quantity is defined as the impulse, $\mathbf{J}$:

$$\vec{\mathbf{J}} = \int_{t_i}^{t_f} \vec{\mathbf{F}} dt.$$

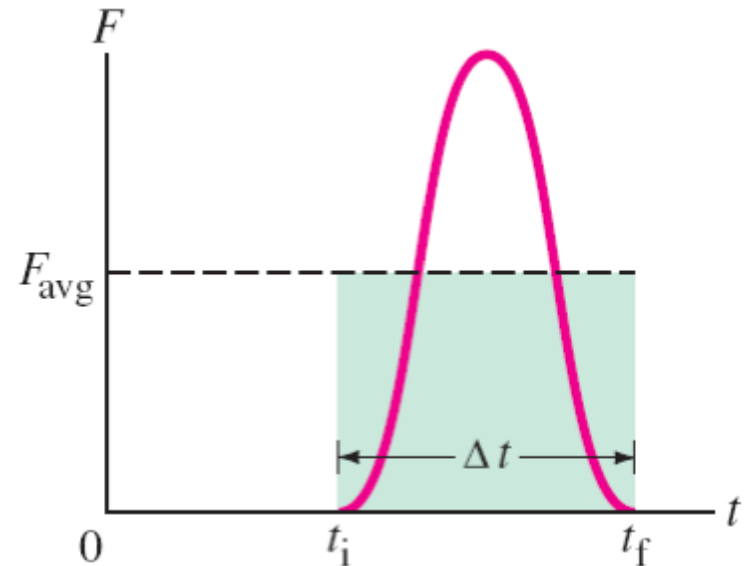
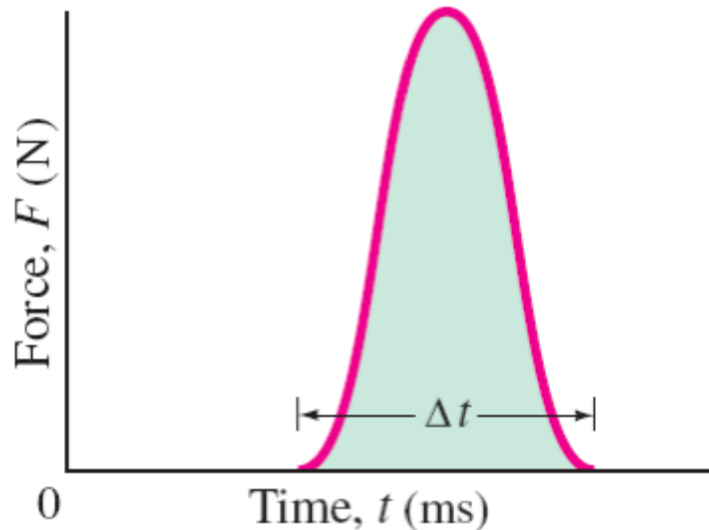
The impulse is equal to the change in momentum:

$$\Delta \vec{\mathbf{p}} = \vec{\mathbf{p}}_f - \vec{\mathbf{p}}_i = \int_{t_i}^{t_f} \vec{\mathbf{F}} dt = \vec{\mathbf{J}}.$$

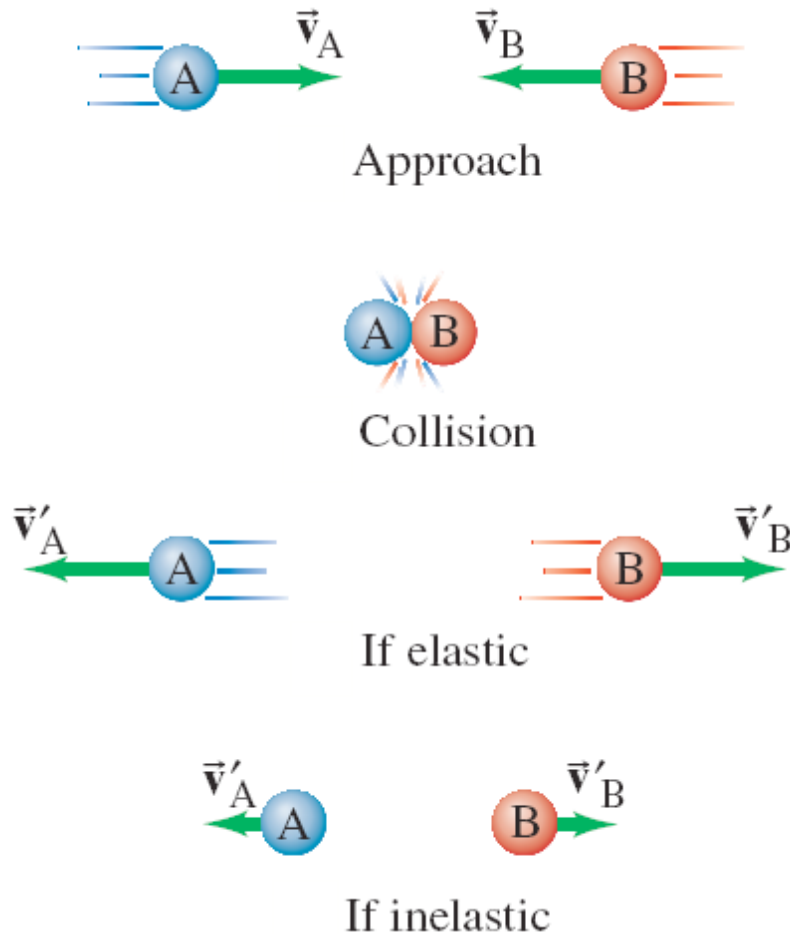
9-3 Collisions and Impulse

Since the **time** of the collision is often very short, we may be able to use the **average force**, which would produce the same impulse over the same time interval.

$$\vec{\mathbf{F}}_{\text{avg}} \Delta t = \int_{t_i}^{t_f} \vec{\mathbf{F}} dt.$$



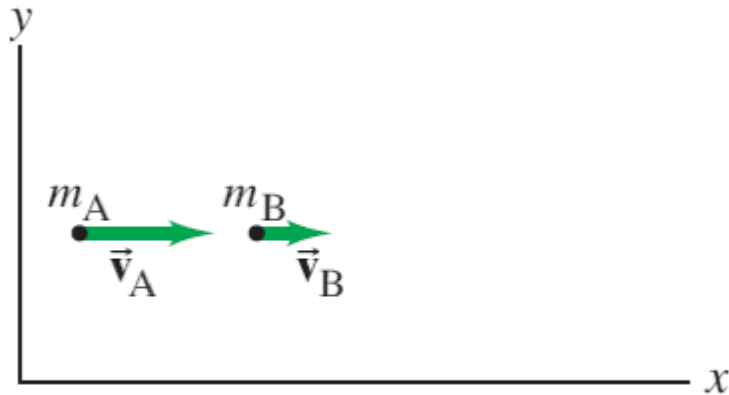
9-4 Conservation of Energy and Momentum in Collisions



Momentum is conserved in all collisions.

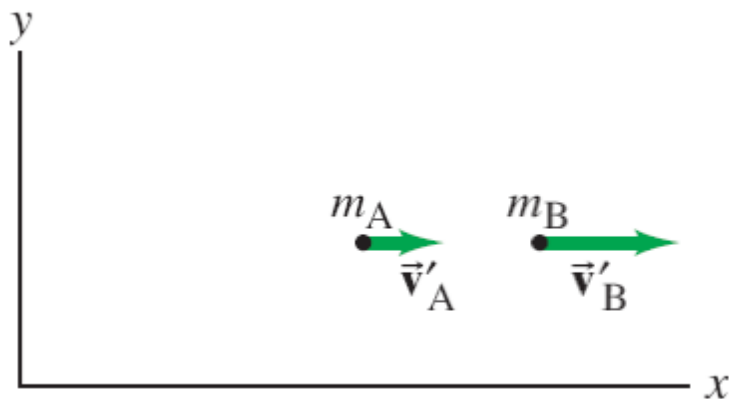
Collisions in which kinetic energy is conserved as well are called elastic collisions, and those in which it is not are called inelastic.

9-5 Elastic Collisions in One Dimension



Here we have two objects colliding **elastically**. We know the masses and the initial speeds.

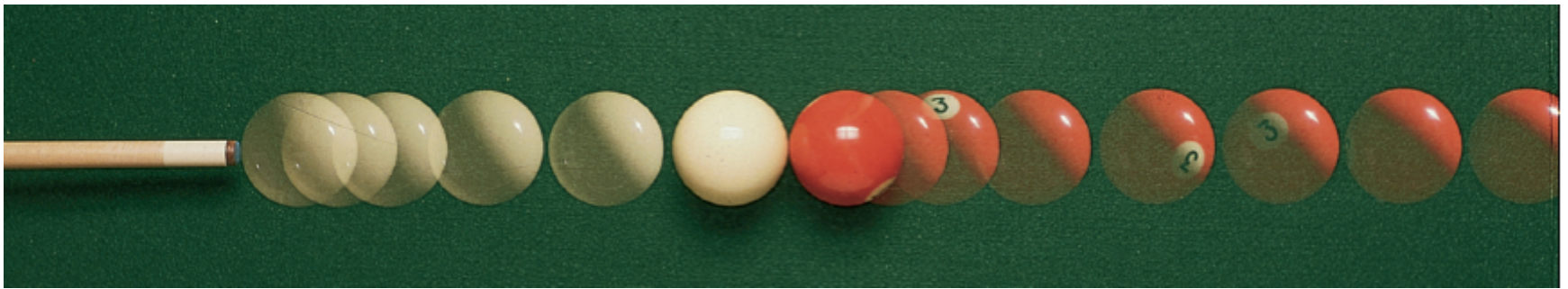
Since both momentum and kinetic energy are conserved, we can write **two** equations. This allows us to solve for the **two** unknown final speeds.



9-5 Elastic Collisions in One Dimension

Example 9-7: Equal masses.

Billiard ball A of mass m moving with speed v_A collides head-on with ball B of equal mass. What are the speeds of the two balls after the collision, assuming it is elastic? Assume (a) both balls are moving initially (v_A and v_B), (b) ball B is initially at rest ($v_B = 0$).



9-5 Elastic Collisions in One Dimension

Example 9-8: Unequal masses, target at rest.

A very common practical situation is for a moving object (m_A) to strike a second object (m_B , the “target”) at rest ($v_B = 0$). Assume the objects have unequal masses, and that the collision is elastic and occurs along a line (head-on). (a) Derive equations for v_B' and v_A' in terms of the initial velocity v_A of mass m_A and the masses m_A and m_B . (b) Determine the final velocities if the moving object is much more massive than the target ($m_A \gg m_B$). (c) Determine the final velocities if the moving object is much less massive than the target ($m_A \ll m_B$).

9-6 Inelastic Collisions

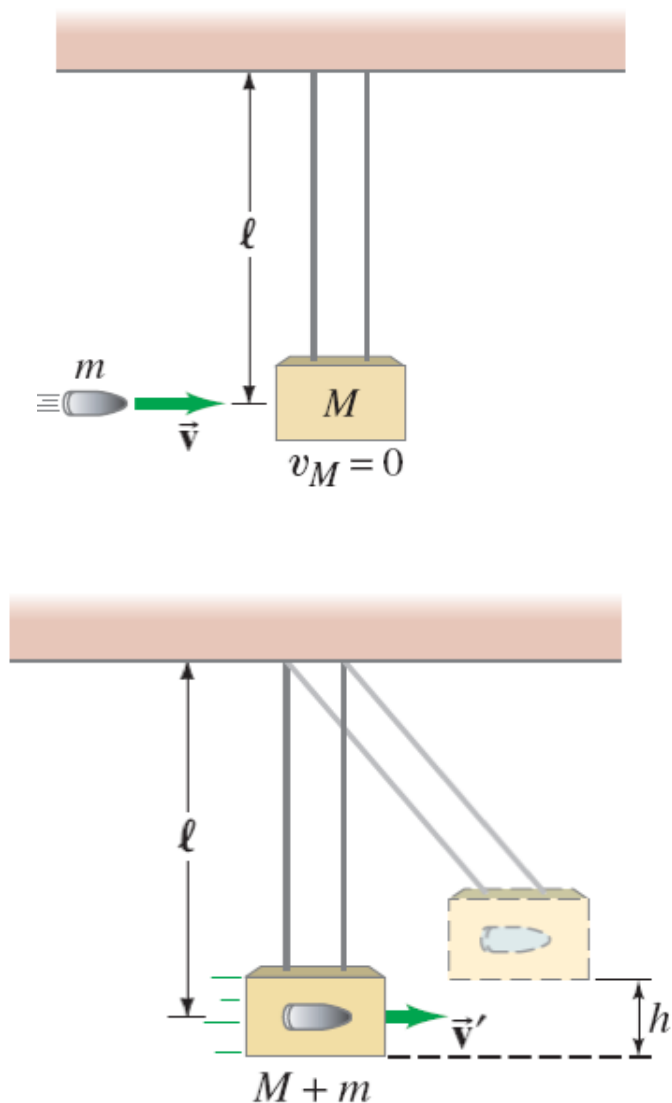
With inelastic collisions, some of the initial kinetic energy is lost to thermal or potential energy. Kinetic energy may also be gained during explosions, as there is the addition of chemical or nuclear energy.

A completely inelastic collision is one in which the objects stick together afterward, so there is only one final velocity.

9-6 Inelastic Collisions

Example 9-11: Ballistic pendulum.

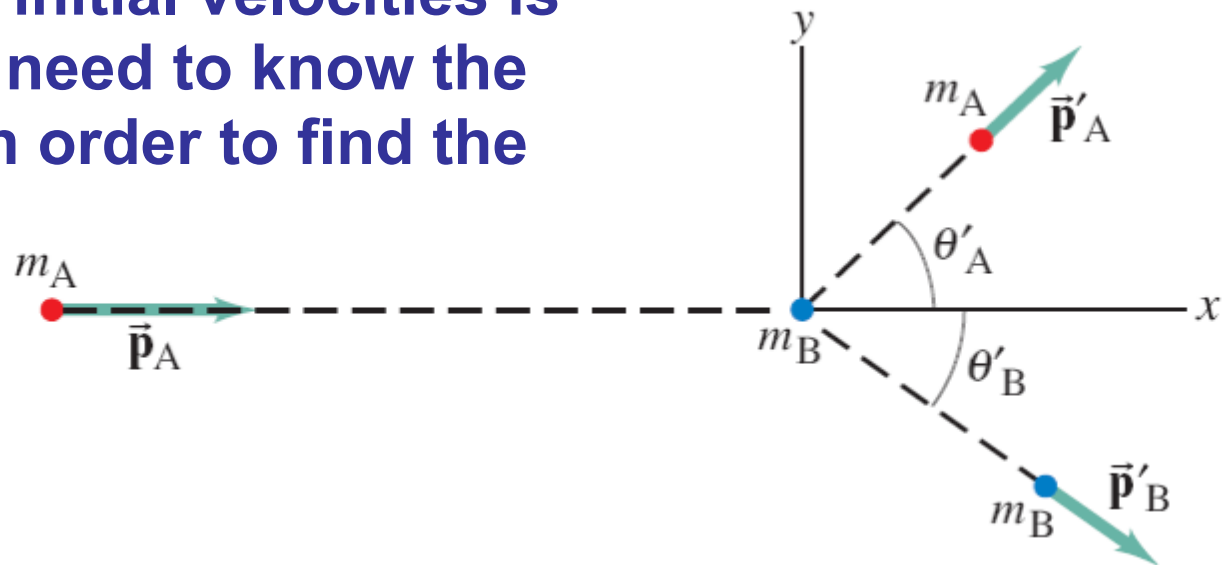
The ballistic pendulum is a device used to measure the speed of a projectile, such as a bullet. The projectile, of mass m , is fired into a large block of mass M , which is suspended like a pendulum. As a result of the collision, the pendulum and projectile together swing up to a maximum height h . Determine the relationship between the initial horizontal speed of the projectile, v , and the maximum height h .



9-7 Collisions in Two or Three Dimensions

Conservation of energy and momentum can also be used to analyze collisions in **two or three** dimensions, but unless the situation is very simple, the math quickly becomes unwieldy.

Here, a **moving** object collides with an object initially at **rest**. Knowing the masses and initial velocities is not enough; we need to know the **angles** as well in order to find the final velocities.



9-7 Collisions in Two or Three Dimensions

Problem solving:

1. Choose the **system**. If it is complex, **subsystems** may be chosen where one or more conservation laws apply.
2. Is there an **external force**? If so, is the **collision time** short enough that you can ignore it?
3. Draw diagrams of the initial and final situations, with **momentum vectors** labeled.
4. Choose a **coordinate system**.

9-7 Collisions in Two or Three Dimensions

5. Apply momentum conservation; there will be one equation for each dimension.

6. If the collision is elastic, apply conservation of kinetic energy as well.

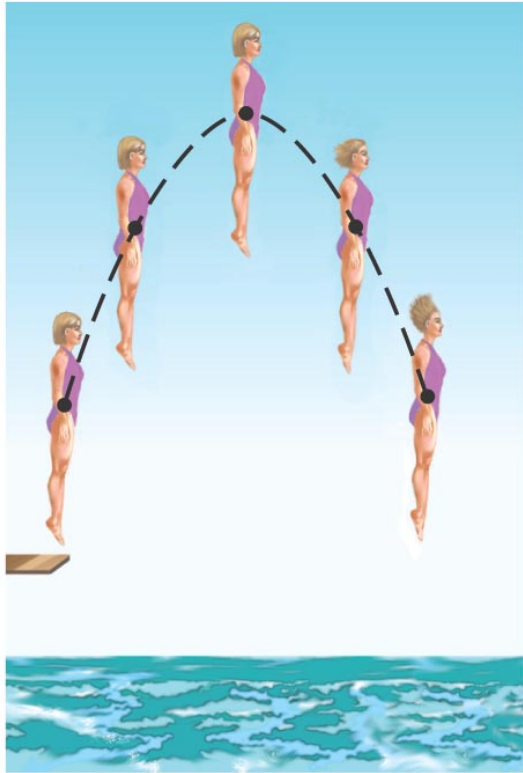
7. Solve.

8. Check units and magnitudes of result.

9-8 Center of Mass (CM)

In (a), the diver's motion is pure translation; in (b) it is translation **plus** rotation.

There is one point that moves in the same path a particle would take if subjected to the same force as the diver. This point is called the **center of mass (CM)**.



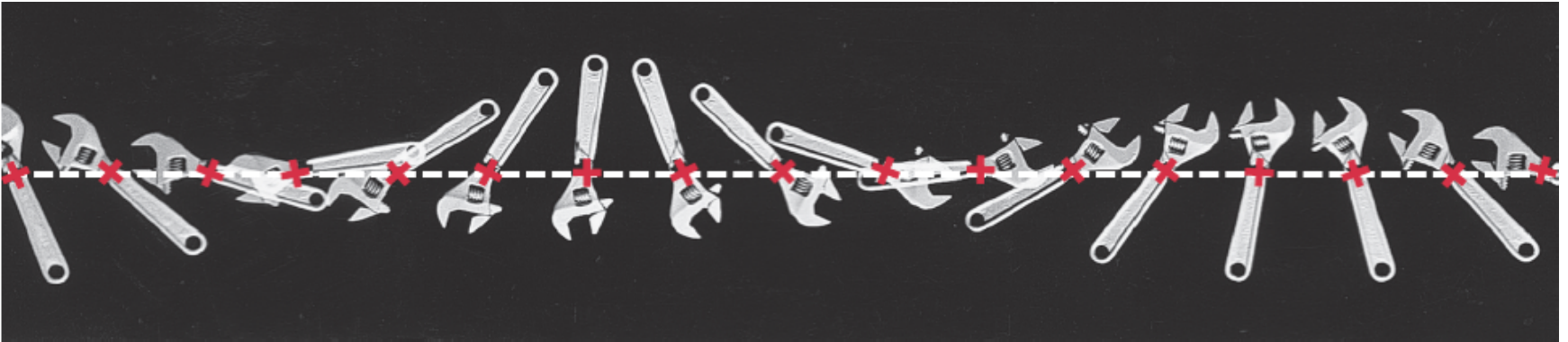
(a)



(b)

9-8 Center of Mass (CM)

The general motion of an object can be considered as the sum of the translational motion of the CM, plus rotational, vibrational, or other forms of motion about the CM.

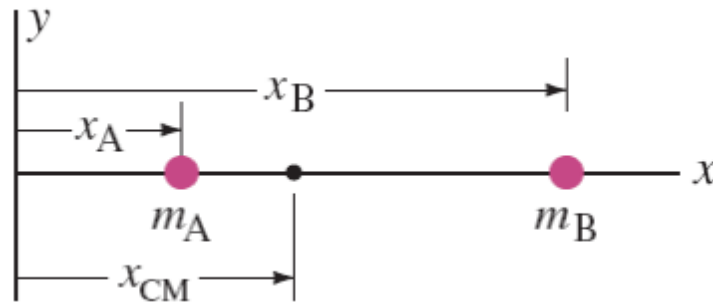


9-8 Center of Mass (CM)

For two particles, the **center of mass lies closer to the one with the most mass:**

$$x_{\text{CM}} = \frac{m_A x_A + m_B x_B}{m_A + m_B} = \frac{m_A x_A + m_B x_B}{M},$$

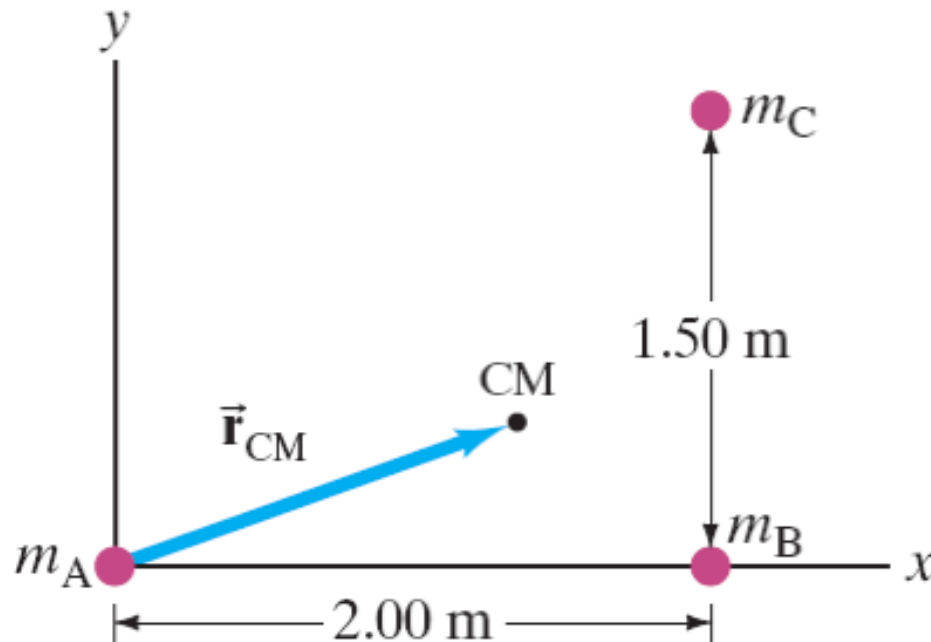
where M is the **total mass**.



9-8 Center of Mass (CM)

Exercise 9-15: Three particles in 2-D.

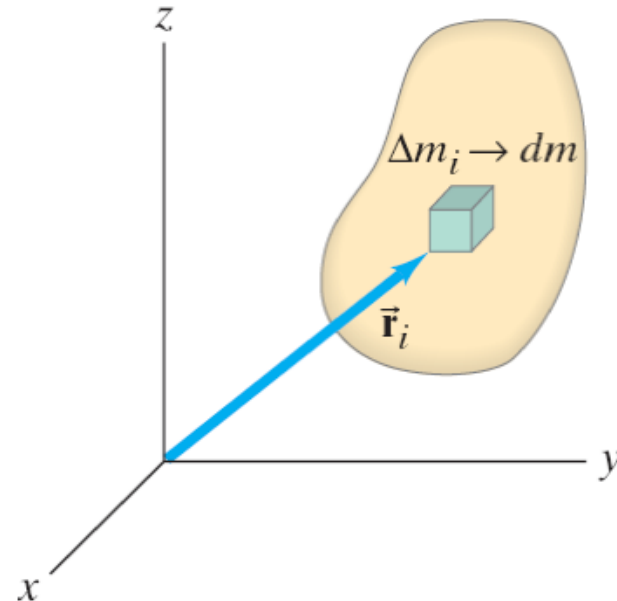
Three particles, each of mass 2.50 kg, are located at the corners of a right triangle whose sides are 2.00 m and 1.50 m long, as shown. Locate the center of mass.



9-8 Center of Mass (CM)

For an extended object, we imagine making it up of tiny particles, each of tiny mass, and adding up the product of each particle's mass with its position and dividing by the total mass. In the limit that the particles become infinitely small, this gives:

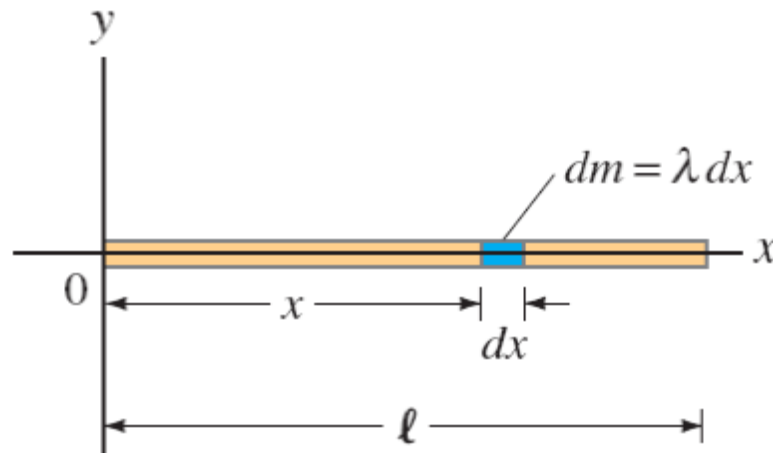
$$\vec{\mathbf{r}}_{\text{CM}} = \frac{1}{M} \int \vec{\mathbf{r}} \, dm.$$



9-8 Center of Mass (CM)

Example 9-16: CM of a thin rod.

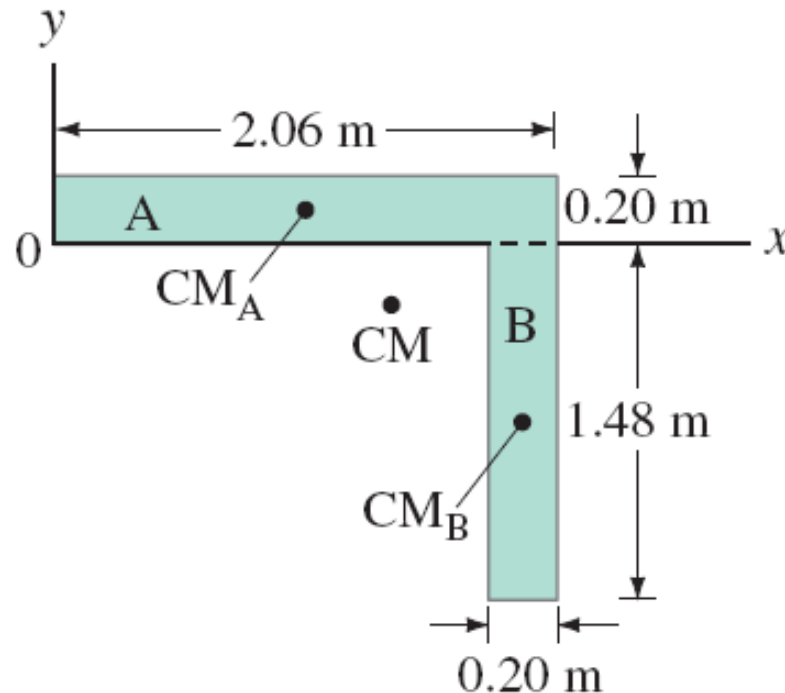
(a) Show that the CM of a uniform thin rod of length l and mass M is at its center. (b) Determine the CM of the rod assuming its linear mass density λ (its mass per unit length) varies linearly from $\lambda = \lambda_0$ at the left end to double that value, $\lambda = 2\lambda_0$, at the right end.



9-8 Center of Mass (CM)

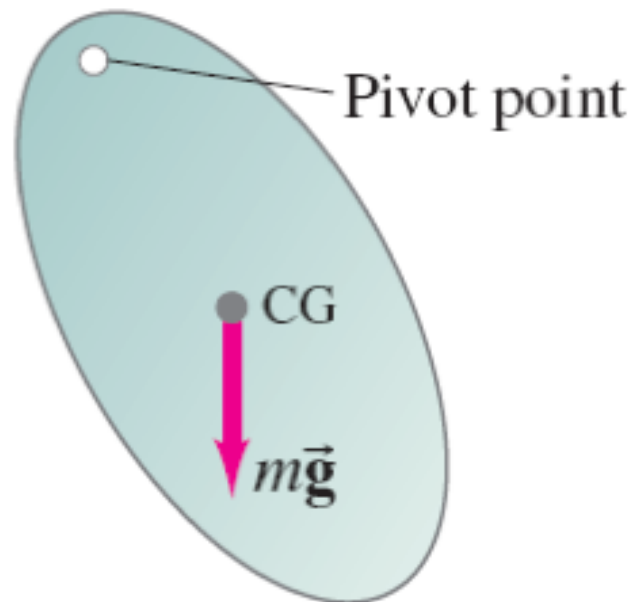
Example 9-17: CM of L-shaped flat object.

Determine the CM of the uniform thin L-shaped construction brace shown.



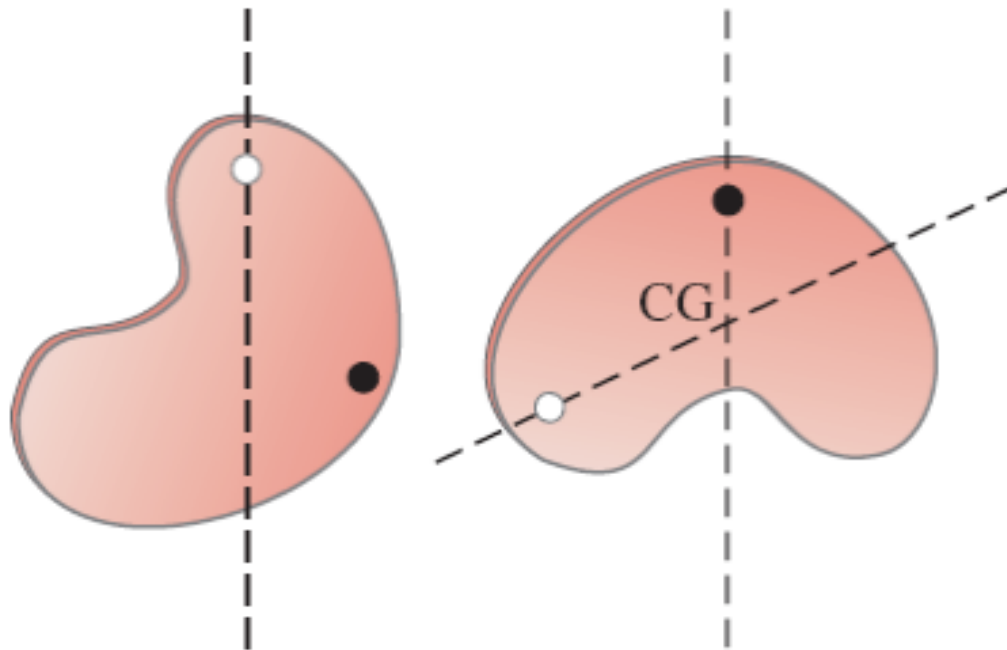
9-8 Center of Mass (CM)

The center of gravity is the point at which the gravitational force can be considered to act. It is the same as the center of mass as long as the gravitational force does not vary among different parts of the object.



9-8 Center of Mass (CM)

The center of gravity can be found experimentally by suspending an object from different points. The CM need not be within the actual object—a doughnut's CM is in the center of the hole.



9-9 Center of Mass and Translational Motion

The total momentum of a system of particles is equal to the product of the total mass and the velocity of the center of mass.

The sum of all the forces acting on a system is equal to the total mass of the system multiplied by the acceleration of the center of mass:

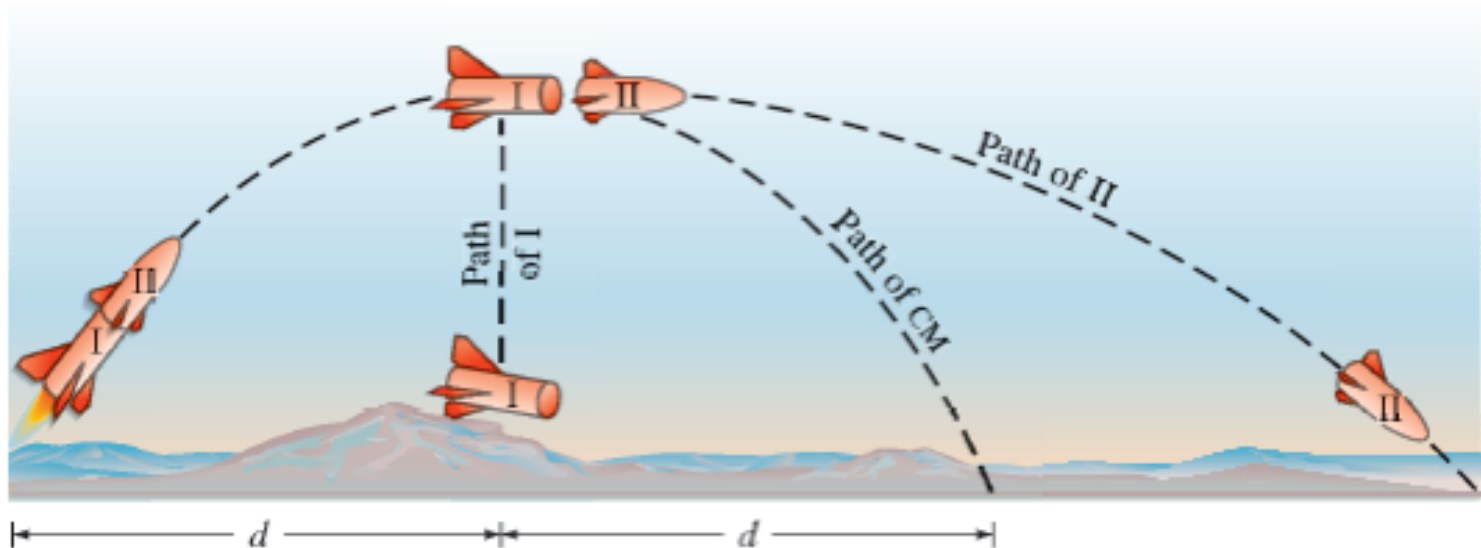
$$M\vec{a}_{\text{CM}} = \Sigma \vec{F}_{\text{ext.}}$$

Therefore, the center of mass of a system of particles (or objects) with total mass M moves like a single particle of mass M acted upon by the same net external force.

9-9 Center of Mass and Translational Motion

Conceptual Example 9-18: A two-stage rocket.

A rocket is shot into the air as shown. At the moment it reaches its highest point, a prearranged explosion separates it into two parts of equal mass. Part I is stopped in midair by the explosion and falls vertically to Earth. Where does part II land? Assume $g = \text{constant}$.



Summary of Chapter 9

- **Momentum of an object:** $\vec{\mathbf{p}} = m\vec{\mathbf{v}}.$
- **Newton's second law:** $\Sigma\vec{\mathbf{F}} = \frac{d\vec{\mathbf{p}}}{dt}.$
- **Total momentum of an isolated system of objects is conserved.**
- **During a collision, the colliding objects can be considered to be an isolated system even if external forces exist, as long as they are not too large.**
- **Momentum will therefore be conserved during collisions.**

Summary of Chapter 9, cont.

- **Impulse:** $\vec{\mathbf{J}} = \int \vec{\mathbf{F}} dt.$
- **In an elastic collision, total kinetic energy is also conserved.**
- **In an inelastic collision, some kinetic energy is lost.**
- **In a completely inelastic collision, the two objects stick together after the collision.**
- **The center of mass of a system is the point at which external forces can be considered to act.**